

THESIS DEFENSE
In Partial Fulfillment of the Requirements for the
Master's Degree in Applied Econometrics

THEME:

What Drives the Cost of Government
Debt in Times of Shifting Policies:
Evidence from Morocco

Jury members:

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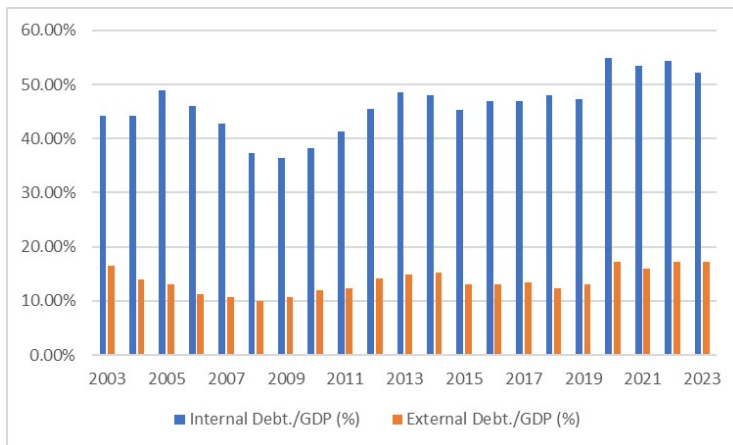
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July 29, 2025

- 1 Context and objectives
- 2 Problem statement
- 3 Related literature
- 4 Methodology and Data
- 5 Results and discussion
- 6 Concluding remarks

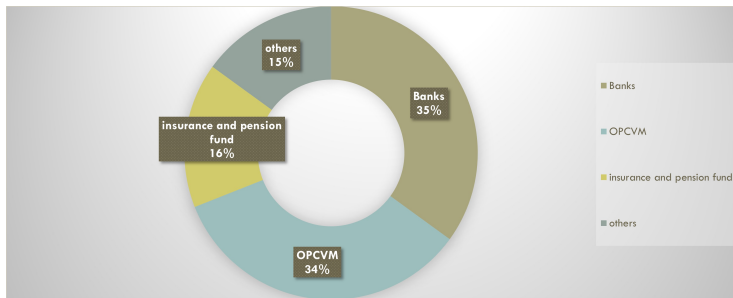
- Following the COVID-19 pandemic, the world entered a period of surging inflation starting in mid-2021.
- Central banks aim to ensure price stability. In response to rising inflation, many increased their policy rates.

Figure 1: Internal and External Debt per GDP



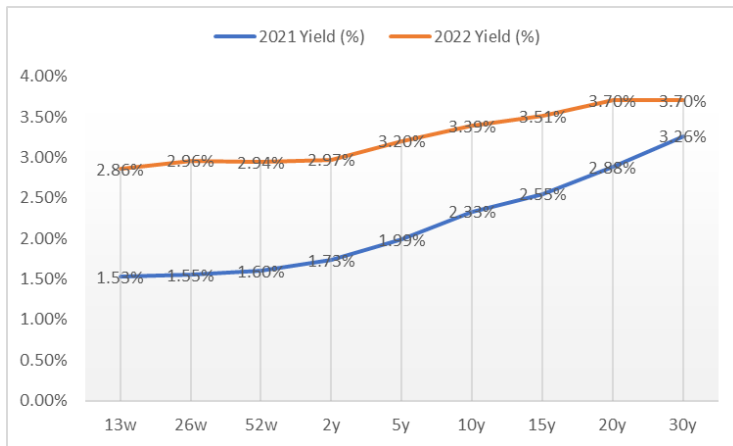
Source: Ministry of Economic and Finance

Figure 2: Holders of Treasury Bonds in 2022



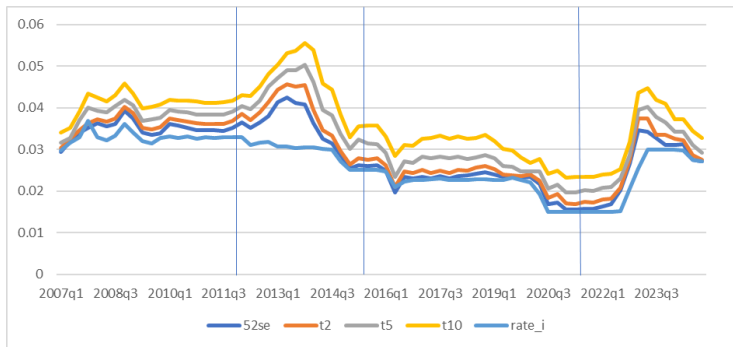
Source: Ministry of Economic and Finance

Figure 3: Yield curve in 2021 and 2022



Source: Authors' elaboration based on Attijari Global Research

Figure 4: Interest rate evolution in Morocco



Source: Bank Al-Maghrib

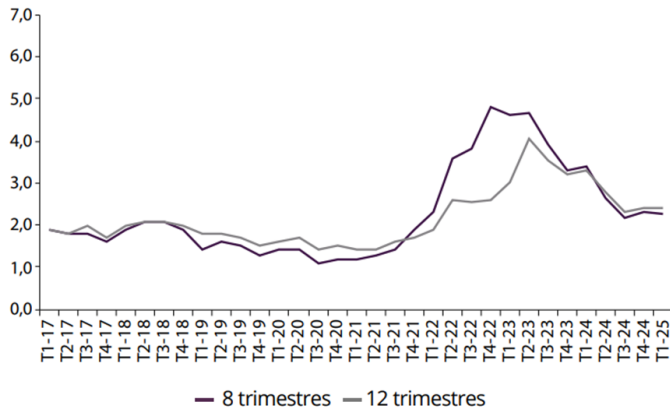
Table 1: Risk Premiums Over Time

Période	Risk premium - 52 week	Risk premium -2 ans	Risk premium -5 ans	Risk premium - 10 ans
2007–2011	20 (bps)	32 (bps)	54 (bps)	82 (bps)
2012–2014	74 (bps)	101 (bps)	140 (bps)	185 (bps)
2015–2021	12 (bps)	25 (bps)	53 (bps)	93 (bps)
2022–2024	21 (bps)	35 (bps)	60 (bps)	95 (bps)

Source: Authors' elaboration

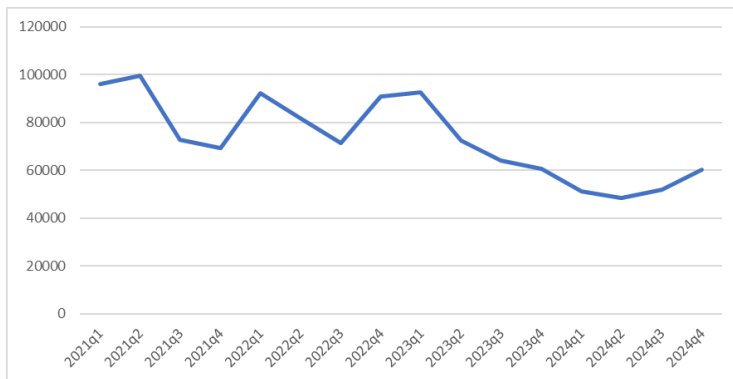
Note: Here, the risk premium is defined as the difference between the key rate (or policy rate) and the corresponding point on the yield curve.

Figure 5: Inflation expectations



Source: Bank Al-Maghrib (Monetary policy report)

Figure 6: Liquidity provided by Bank Al-Maghrib



Source: Bank Al-Maghrib

Problem statement: What is the impact of Monetary Tightening on the Cost of Sovereign Debt in the Post-COVID era?

Table 2: Theoretical Insights

Authors	Year	Transmission Channel	Results
Rolley & Sellon	1995	Interest rate	Relationship between short- and long-term rates varies across economic cycles. Restrictive policy during upward cycle → increase in long-term rates. End of cycle → Limited impact on long-term rates due to policy anticipation.
Poole	2006	Inflation expectations	Risk premium fluctuates over time, affecting how long-term rates respond to policy shifts.
Lowe	1995	Liquidity channel	Lower short-term rates → Increased long-term bond purchases by investors → Reduced long-term rates. Inflation expectations: Monetary easing → Anticipation of inflation → Higher long-term rates.
Heider et al.	2009	Liquidity channel	Increased counterparty risk → Increased liquidity hoarding by banks → Higher interbank rates (short maturities) → Flattening of the yield curve.

Source: Authors' elaboration

Table 3: Empirical Findings

Authors	Transmission Channel	Method / Focus	Results
Cook & Hahn (1989)	Interest rate	Simple regression	Changes in the policy rate lead to: - Significant movement in short-term rates - Moderate response in medium-term rates - Limited effect on long-term rates
Bulř & Vlček (2016), IMF WP	Interest rate	SVAR models on panel of 16 countries (developed & emerging)	Developed countries: Yield curve flattens post-policy shock due to central bank credibility. Developing countries: Yield curve shifts upward due to weak inflation anchoring.
Kwan Soo Bong, Taeyoung Doh, & Woong Yong Park (2016)	Interest rate	Bayesian DSGE model	Long-term rates are highly correlated with both short-term and expected short-term rates.

Source: Authors' elaboration based on original studies.

Table 4: Empirical Findings

Authors	Focus / Country	Transmission Channel	Main Results
Bernanke & Blinder (1992)	General (US)	Liquidity channel	Monetary policy affects bank loan supply through: – Mechanism 1: Tightening reduces deposit returns → less loanable funds. – Mechanism 2: Higher cost of market funding → increased external financing premium → reduced lending.
Bonam, Ciccarelli & Gomes (2024)	Post-COVID-19	Interest rate	Higher interest rates increase debt service burden → reduced fiscal space. If $r - g > 0$, fiscal sustainability worsens.
Ahokpossi et al. (2016)	Morocco	Interest rate	Tighter monetary policy leads to higher short-term interest rates.
Bennouna & Bounader (2018)	Morocco	Interest rate	Yield curve steepening observed. Long-term rates are more sensitive to monetary policy than short-term rates.

Source: Authors' elaboration

Our research contributes to the literature in three key ways:

- i. by emphasizing the credit channel of monetary policy,
- ii. by adopting a two methodological framework in the empirical analysis,
- iii. by focusing on the post-COVID-19 period, which is characterized by an unusual combination of monetary tightening and expansionary fiscal policy.

- The empirical approach to study the transmission adopted in this study is inspired by the work of **Christiano-Eichenbaum-Evans (1996)**, who use Structural Vector Autoregressive models.
- Why this method matter:
 - Unlike Dynamic Stochastic General Equilibrium (DSGE) models, SVAR models offer a more flexible and empirically grounded alternative that allows us to simulate the effects of monetary policy shocks.
 - Structural VAR is just identified, unlike the traditional simultaneous equations that often impose more restrictions than necessary, leading to over-identification.

To simulate the transmission of monetary policy (interest rate channel) shocks on the yield curve, our model includes the following macroeconomic variables:

$$t52_t = \alpha_0 + a_1 t2_{t-1} + a_2 GDP_{t-1} + a_3 CPI_{t-1} + a_4 IRFCL_{t-1} + a_5 Spread_t52_{t-1} + a_6 rate_i_{t-1} + \rho_t \quad (1)$$

$$t2_t = \theta_0 + \theta_1 t2_{t-1} + \theta_2 GDP_{t-1} + \theta_3 CPI_{t-1} + \theta_4 IRFCL_{t-1} + \theta_5 Spread_t2_{t-1} + \theta_6 rate_i_{t-1} + \omega_t \quad (2)$$

$$t5_t = \beta_0 + \beta_1 t5_{t-1} + \beta_2 GDP_{t-1} + \beta_3 CPI_{t-1} + \beta_4 IRFCL_{t-1} + \beta_5 Spread_t2_{t-1} + \beta_6 rate_i_{t-1} + \mu_t \quad (3)$$

$$t10_t = \gamma_0 + \gamma_1 t10_{t-1} + \gamma_2 GDP_{t-1} + \gamma_3 CPI_{t-1} + \gamma_4 IRFCL_{t-1} + \gamma_5 Spread_t2_{t-1} + \gamma_6 rate_i_{t-1} + \epsilon_t \quad (4)$$

Also, to simulate the transmission of monetary policy (Credit channel) shocks on the yield curve, our model includes the following macroeconomic variables:

$$t52_t = \alpha_0 + a_1 t2_{t-1} + a_2 GDP_{t-1} + a_3 CPI_{t-1} + a_4 IRFCL_{t-1} + a_5 Spread_t52_{t-1} + a_6 Volume_jj_{t-1} + \rho_t \quad (5)$$

$$t2_t = \theta_0 + \theta_1 t2_{t-1} + \theta_2 GDP_{t-1} + \theta_3 CPI_{t-1} + \theta_4 IRFCL_{t-1} + \theta_5 Spread_t2_{t-1} + \theta_6 Volume_jj_{t-1} + \omega_t \quad (6)$$

$$t5_t = \beta_0 + \beta_1 t5_{t-1} + \beta_2 GDP_{t-1} + \beta_3 CPI_{t-1} + \beta_4 IRFCL_{t-1} + \beta_5 Spread_t2_{t-1} + \beta_6 Volume_jj_{t-1} + \mu_t \quad (7)$$

$$t10_t = \gamma_0 + \gamma_1 t10_{t-1} + \gamma_2 GDP_{t-1} + \gamma_3 CPI_{t-1} + \gamma_4 IRFCL_{t-1} + \gamma_5 Spread_t2_{t-1} + \gamma_6 Volume_jj_{t-1} + \epsilon_t \quad (8)$$

Structural Vector Autoregressive methodology

The structural form of a VAR model is presented as follows:

$$\begin{bmatrix} Y_t \\ R_t^j \end{bmatrix} = \begin{bmatrix} A(L) & B(L) \\ C(L) & D(L) \end{bmatrix} \begin{bmatrix} Y_{t-1} \\ R_{t-1}^j \end{bmatrix} + \begin{bmatrix} \varepsilon_t^Y \\ \varepsilon_t^j \end{bmatrix} \quad (9)$$

Where:

R_t^j : Yield curve component for each maturity j

Y_t : Vector of endogenous variables such that:

$$Y_t \equiv (GDP_t, CPI_t, spread_t, IRFCL_t, rate_i \text{ or } Volume_jj_t)$$

Identification strategy

Following Sims (1980), one common identification strategy is to apply a Cholesky decomposition to the reduced-form covariance matrix. This identification strategy involves imposing the following restrictions:

$$\varepsilon_t^y = \begin{bmatrix} \varepsilon_t^{GDP} \\ \varepsilon_t^{CPI} \\ \varepsilon_t^{spread_n} \\ \varepsilon_t^{IRFCL} \\ \varepsilon_t^{rate_i \text{ or } Volume_jj_t} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ a_{21} & 1 & 0 & 0 & 0 \\ a_{31} & a_{32} & 1 & 0 & 0 \\ a_{41} & a_{42} & a_{43} & 1 & 0 \\ a_{51} & a_{52} & a_{53} & a_{54} & 1 \end{bmatrix} \begin{bmatrix} u_t^{GDP} \\ u_t^{CPI} \\ u_t^{spread_n} \\ u_t^{IRFCL} \\ u_t^{rate_i \text{ or } Volume_jj_t} \end{bmatrix} \quad (10)$$

Identification strategy

The rationale behind the Cholesky identification strategy is:

- The first variable in the ordering is treated as the most exogenous — it affects all others contemporaneously.
- The last variable is the most endogenous — it is affected contemporaneously by all others but does not affect them contemporaneously.

This recursive identification relies on two key assumptions:

- 1 The central bank has contemporaneous access to all relevant information regarding economic activity and inflationary pressures when setting the key rate.
- 2 Monetary policy decisions transmit with a delay to the broader economy.

Local projections methodology

Local projections, introduced by Jordà (2005), estimate impulse responses by directly regressing future outcomes on current variables — one forecast horizon at a time.

The general form of the model is:

$$R_{t+h}^j = \alpha_h + \beta_h \cdot InterBR_t + \gamma_h \cdot Z_t + \varepsilon_{t+h} \quad (11)$$

Where:

- R_{t+h}^j : Endogenous variable (e.g., yield) at horizon h .
- $InterBR_t$: Shock or variable of interest (interbank rate).
- Z_t : Control variables including GDP, CPI, spread (MAD vs. US), and IRFCL.

For each horizon $h = 1, 2, \dots, H$, a separate linear regression is estimated. The objective is to estimate β_h , which captures the dynamic response over time.

Table 5: Variable definitions and sources

Variables	Definition	Source
Interbank rate (rate_i)	Overnight interbank market rate. It is a weighted average rate based on the amounts of reported transactions for a representative sample of institutions admitted to the interbank market.	Bank AL-Maghrib
Liquidity volume (volume_jj)	Volume of liquidity provided by Bank AL-Maghrib on the overnight money market.	Bank AL-Maghrib
GDP	Logarithm of GDP.	Bank AL-Maghrib
CPI	Logarithm of the Consumer Price Index (CPI).	International Monetary Fund
Moroccan 52 Week Bill Yield (t-52)	Interest rate of 52-week maturity treasury bonds.	Bank AL-Maghrib
Morocco 2-Year Bond Yield (t-2)	Interest rate of 2-year maturity treasury bonds.	Bank AL-Maghrib
Morocco 5-Year Bond Yield (t-5)	Interest rate of 5-year maturity treasury bonds.	Bank AL-Maghrib
Morocco 10-Year Bond Yield (t-10)	Interest rate of 10-year maturity treasury bonds.	Bank AL-Maghrib

Source: Authors' elaboration

Table 6: Variable definitions and sources

Variables	Definition	Source
Spread between MAD-US 52-week bills (spread_t52)	This variable represents the risk premium associated with 52-week treasury bills. It is constructed by calculating the difference between the yields on Moroccan 52-week treasury bills and U.S. 52-week treasury bills.	U.S. 52-week treasury bill data from the Federal Reserve Bank of St. Louis (FRED)
Spread between MAD-US 2-year bond yield (spread_t2)	This variable represents the risk premium associated with 2-year treasury bills. It is constructed by calculating the difference between the yields on Moroccan 2-year treasury bills and U.S. 2-year treasury bills.	U.S. 2-year treasury bill data from the Federal Reserve Bank of St. Louis (FRED)

Source: Authors' elaboration

Table 7: Variable definitions and sources

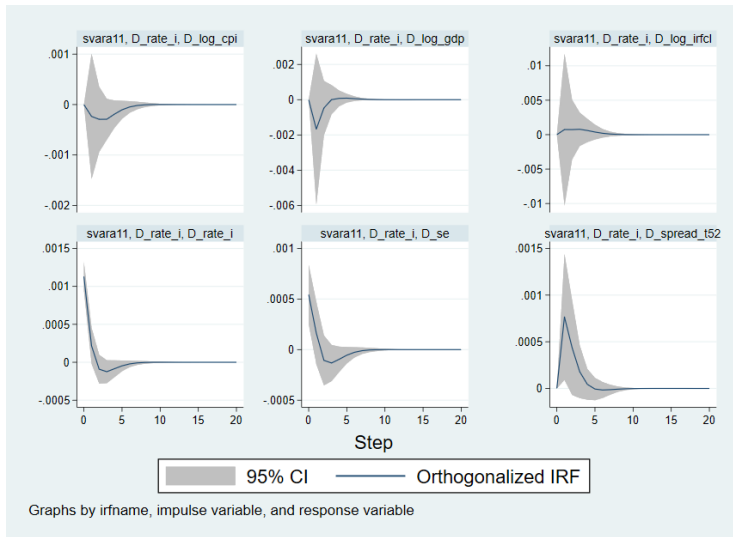
Variables	Definition	Source
Spread between MAD-US 5-year bond yield (spread_t5)	This variable represents the risk premium associated with 5-year treasury bills. It is constructed by calculating the difference between the yields on Moroccan 5-year treasury bills and U.S. 5-year treasury bills.	U.S. 5-year treasury bill data from the Federal Reserve Bank of St. Louis (FRED)
Spread between MAD-US 10-year bond yield (spread_t10)	This variable represents the risk premium associated with 10-year treasury bills. It is constructed by calculating the difference between the yields on Moroccan 10-year treasury bills and U.S. 10-year treasury bills.	U.S. 10-year treasury bill data from the Federal Reserve Bank of St. Louis (FRED)
Reserve assets (IRFCL)	This variable represents the reserve assets that are external and controlled by the monetary authorities for meeting balance of payments financing needs, currency market intervention, and other related purposes.	International Monetary Fund

Source: Authors' elaboration

- We compute the IRFs to simulate a shock of one SD that represent 17 basis point in the interbank rate.
- The variable labelled first is the impulse variable and the variable labelled second is the response variable. All Figures include 20 quarters, equal to five years.

Interest rate channel: SVAR methodology

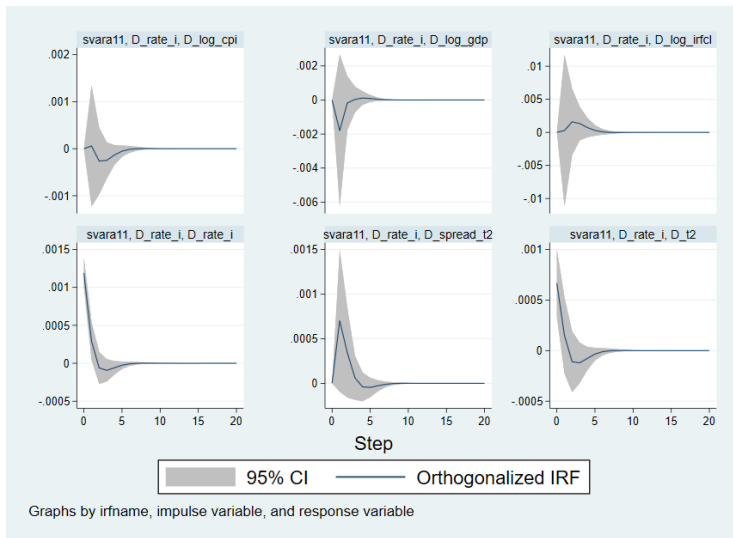
Figure 7: Response of the 52-Week Treasury Bond to a Monetary Policy Shock – SVAR Methodology



Source: Authors' calculation

Interest rate channel: SVAR methodology

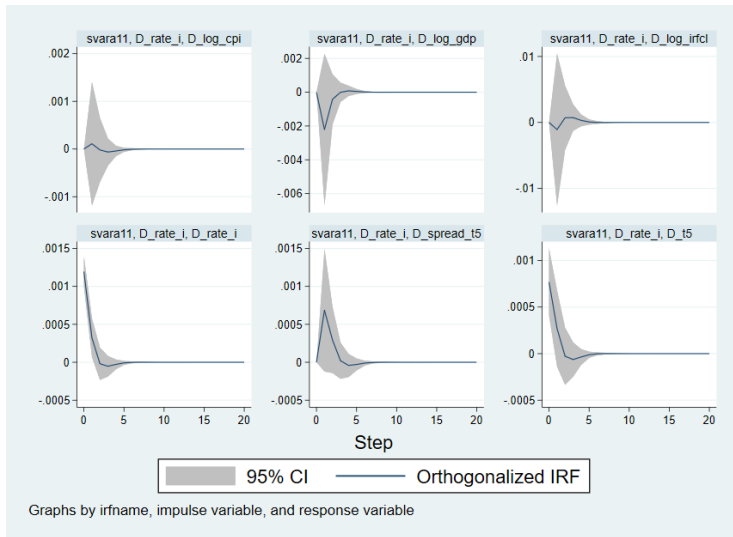
Figure 8: Response of the 2-Year Treasury Bond to a Monetary Policy Shock – SVAR Methodology



Source: Authors' calculation

Interest rate channel: SVAR methodology

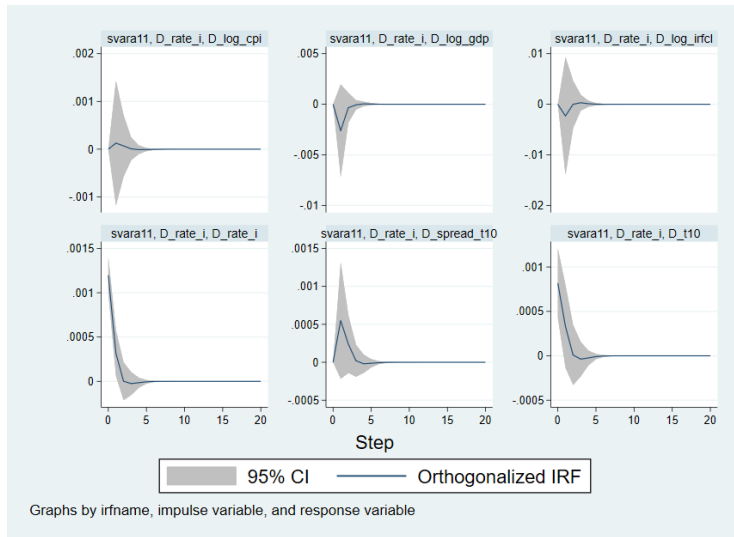
Figure 9: Response of the 5-Year Treasury Bond to a Monetary Policy Shock – SVAR Methodology



Source: Authors' calculation

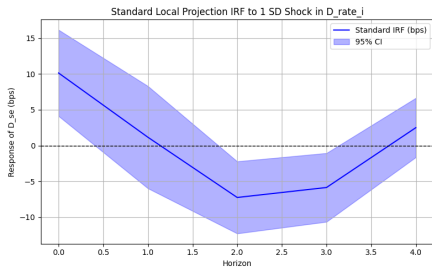
Interest rate channel: SVAR methodology

Figure 10: Response of the 10-Year Treasury Bond to a Monetary Policy Shock – SVAR Methodology

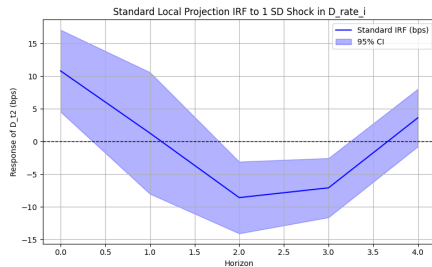


Source: Authors' calculation

Figure 11: Response of the Treasury Bond to a Monetary Policy Shock – LPs Methodology



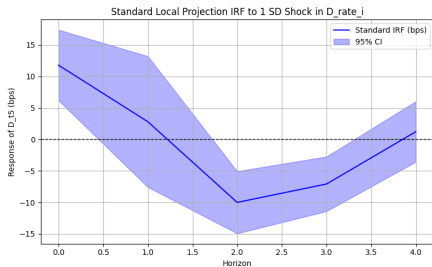
52-week maturity



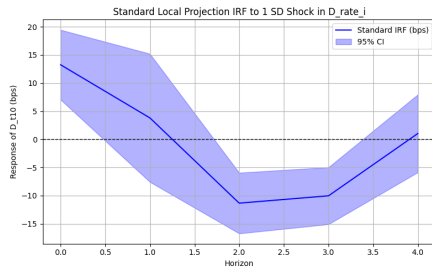
2-year maturity

Source: Authors' calculation

Figure 12: Response of the Treasury Bond to a Monetary Policy Shock – LPs Methodology



5-year maturity



10-year maturity

Source: Authors' calculation

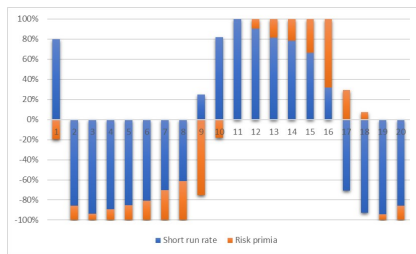
Decomposing the Effect of Monetary Policy Across Different Maturities

- Said previously, we can split the effect of the interest rate channel into two components:
 - Key rate;
 - Risk premium.
- To decompose these effects, we adopt the methodology of Evans and Marshall (1998).
- The risk premium is calculated as the difference between the impulse response of long-term yields and the average of short-term interbank rates over the initial periods, using the following formula:

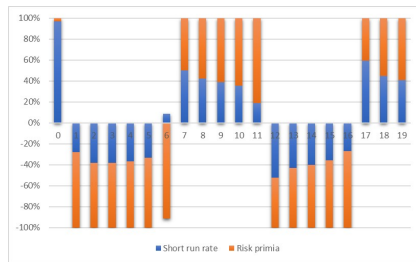
$$\text{Risk Premium}_t^j = \text{Yield}_t^j - \frac{1}{j} \sum_{i=0}^{j-1} \mathbb{E}_t [\text{Rate}_{t+i}] \quad (12)$$

Decomposing the Effect of Monetary Policy Across Different Maturities

Figure 13: Decomposition of the yield curve response to the Monetary Policy shock



52-week maturity

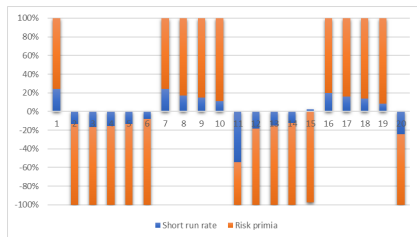


2-year maturity

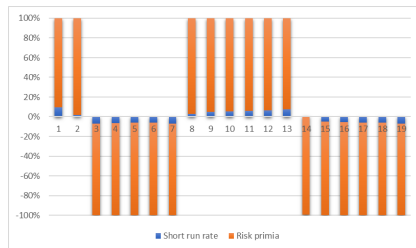
Source: Authors' calculation

Decomposing the Effect of Monetary Policy Across Different Maturities

Figure 14: Decomposition of the yield curve response to the Monetary Policy shock



5-year maturity

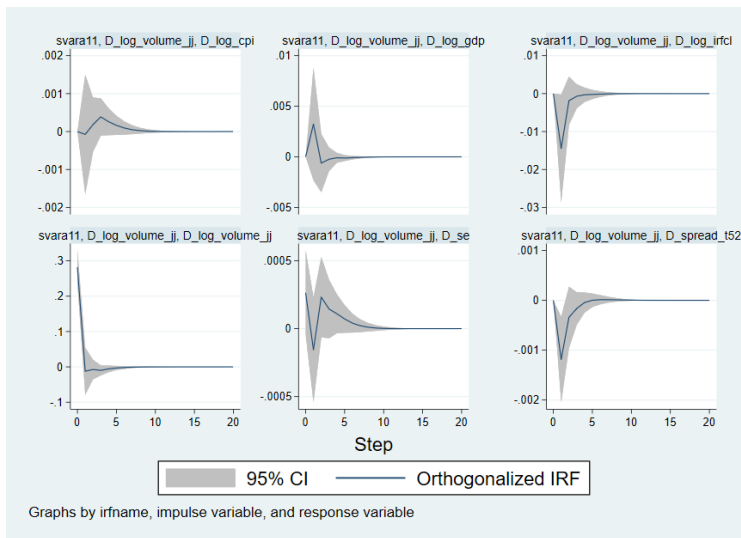


10-year maturity

Source: Authors' calculation

Credit channel: Examining Bank Al-Maghib's liquidity hicks

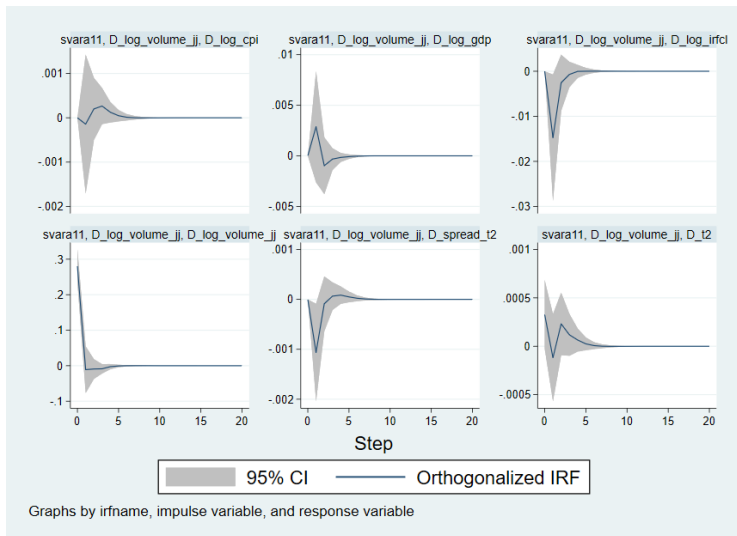
Figure 15: Response of the 52-Week Treasury Bond to a Monetary Policy Shock – SVAR Methodology



Source: Authors' calculation

Credit channel: Examining Bank Al-Maghib's liquidity hicks

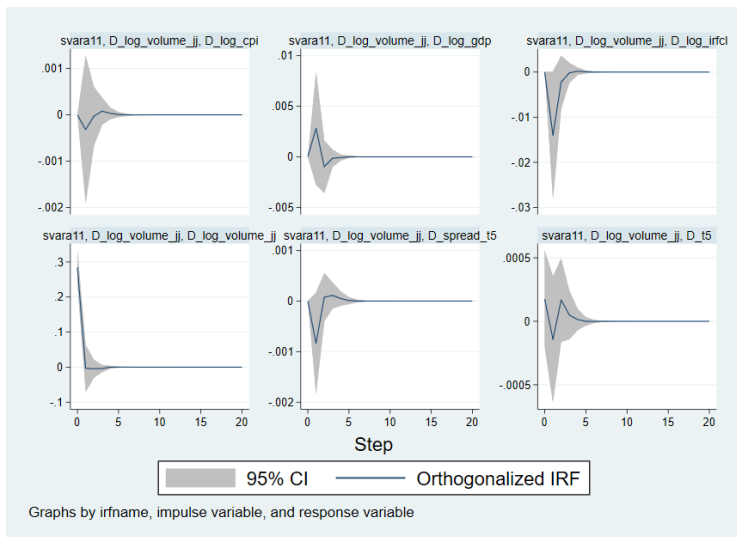
Figure 16: Response of the 2-Year Treasury Bond to a Monetary Policy Shock – SVAR Methodology



Source: Authors' calculation

Credit channel: Examining Bank Al-Maghib's liquidity hicks

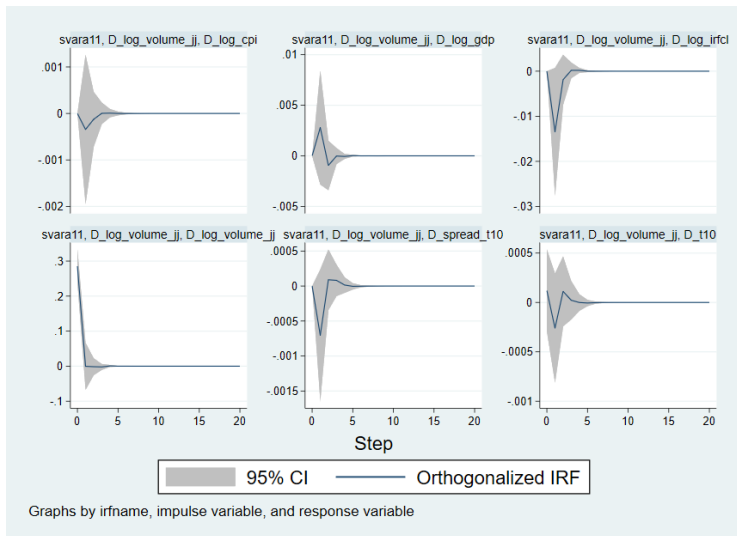
Figure 17: Response of the 5-Year Treasury Bond to a Monetary Policy Shock – SVAR Methodology



Source: Authors' calculation

Credit channel: Examining Bank Al-Maghib's liquidity hicks

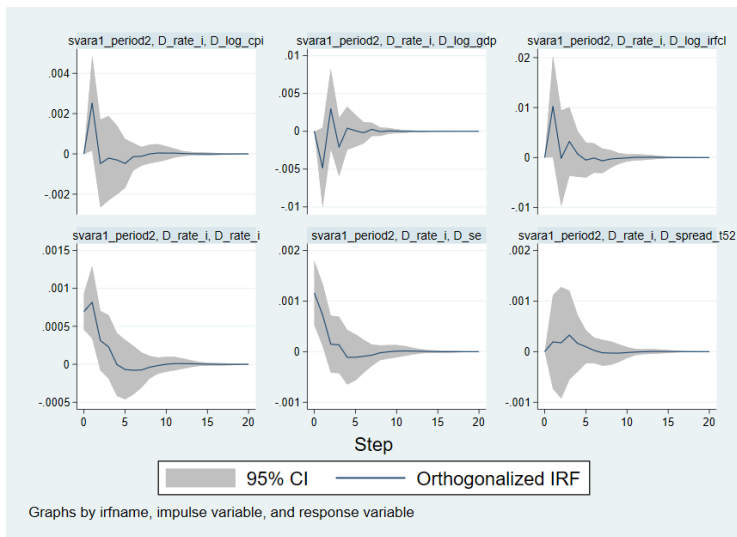
Figure 18: Response of the 10-Year Treasury Bond to a Monetary Policy Shock – SVAR Methodology



Source: Authors' calculation

Post-covid dynamics: Examining the MP shock- Interest rate channel

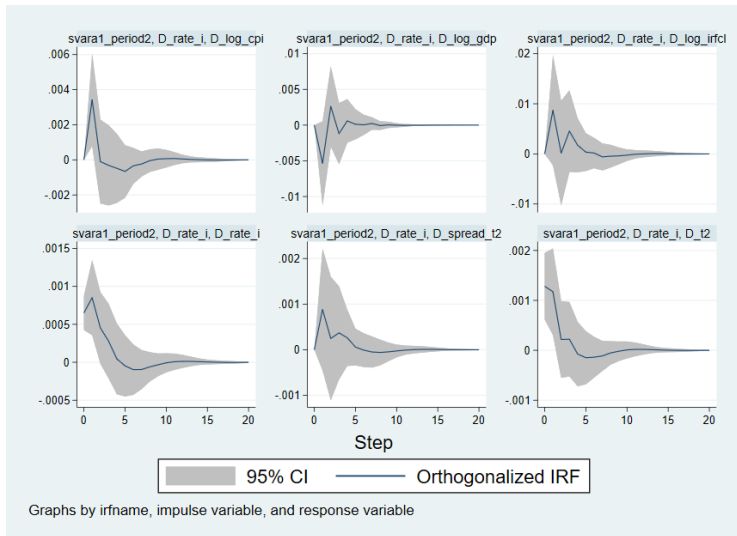
Figure 19: Response of the 52-Week Treasury Bond to a Monetary Policy Shock – Post-covid era



Source: Authors' calculation

Post-covid dynamics: Examining the MP shock - Interest rate channel

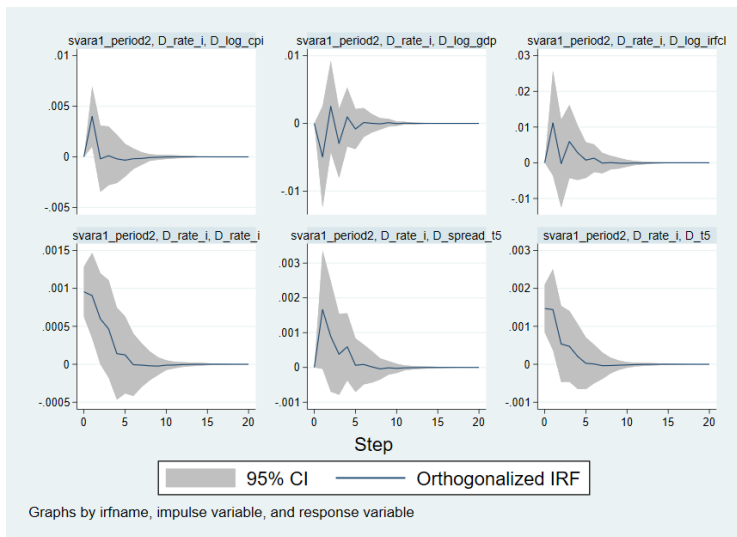
Figure 20: Response of the 2-Year Treasury Bond to a Monetary Policy Shock – Post-covid era



Source: Authors' calculation

Post-covid dynamics: Examining the MP shock - Interest rate channel

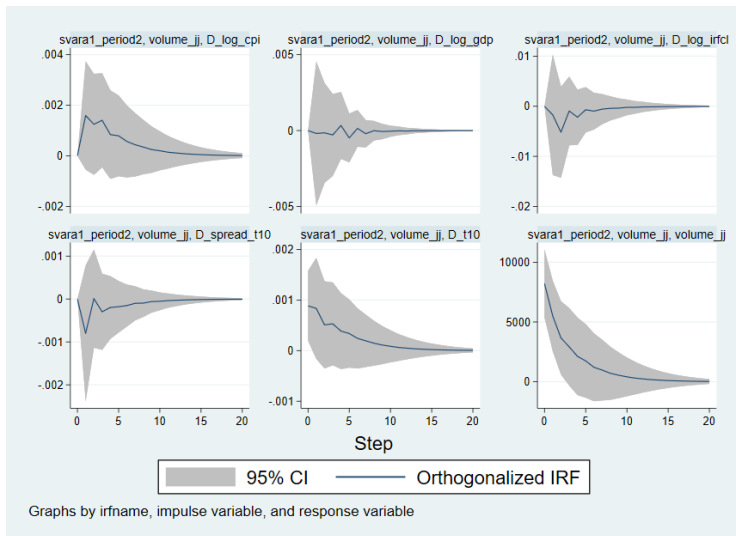
Figure 21: Response of the 5-Year Treasury Bond to a Monetary Policy Shock – Post-covid era



Source: Authors' calculation

Post-covid dynamics: Examining the MP shock - Interest rate channel

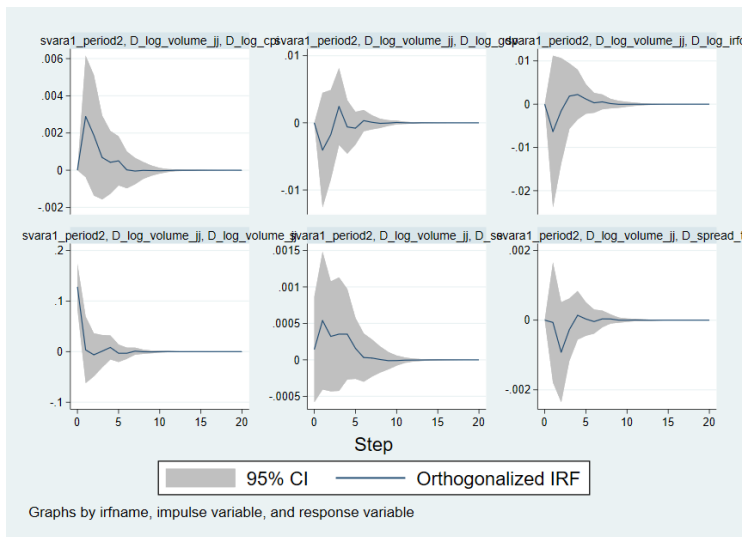
Figure 22: Response of the 10-Year Treasury Bond to a Monetary Policy Shock – Post-covid era



Source: Authors' calculation

Post-covid dynamics: Examining the MP shock - Credit channel

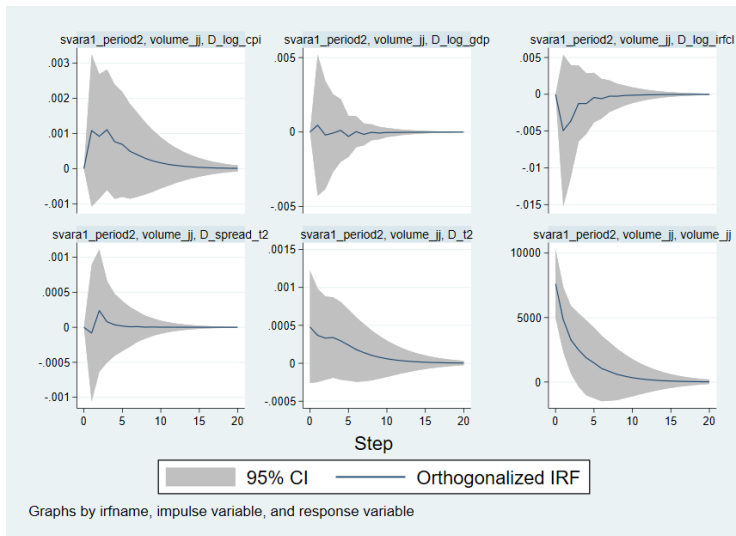
Figure 23: Response of the 52-Week Treasury Bond to a Monetary Policy Shock – Post-covid era



Source: Authors' calculation

Post-covid dynamics: Examining the MP shock - Credit channel

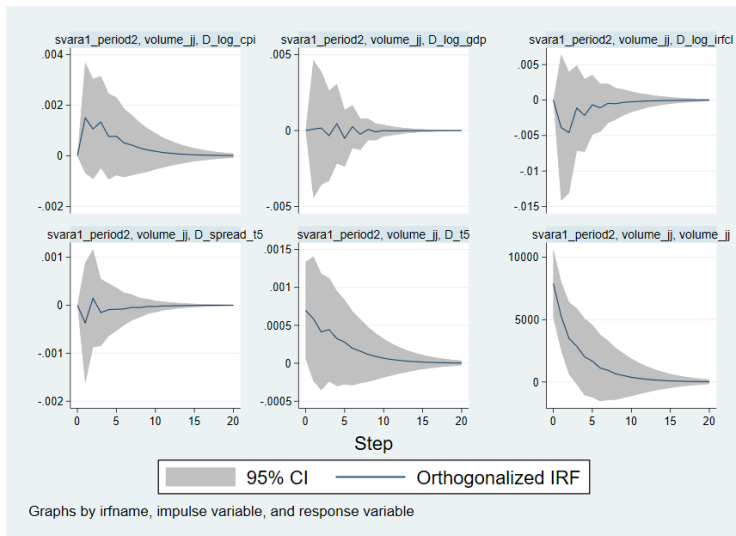
Figure 24: Response of the 2-Year Treasury Bond to a Monetary Policy Shock – Post-covid era



Source: Authors' calculation

Post-covid dynamics: Examining the MP shock - Credit channel

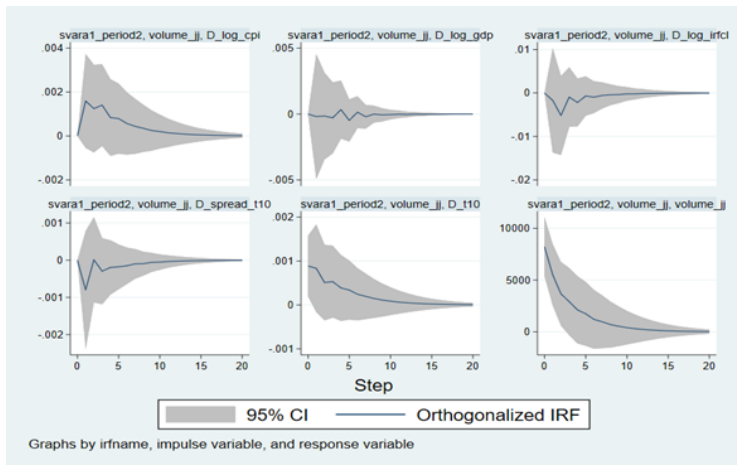
Figure 25: Response of the 5-Year Treasury Bond to a Monetary Policy Shock – Post-covid era



Source: Authors' calculation

Post-covid dynamics: Examining the MP shock - Credit channel

Figure 26: Response of the 10-Year Treasury Bond to a Monetary Policy Shock – Post-covid era



Source: Authors' calculation

- Monetary policy decisions by Bank Al-Maghrib have a significant impact on the yield curve, particularly at the 5-year and 10-year maturities.
- Increases in the policy rate lead to higher yields, as investors anticipate tighter monetary conditions and demand a higher risk premium.
- The yield spread between the United States and Morocco reflects investor concerns about domestic inflation and macroeconomic risks.

- **Main determinants of sovereign bond yields:**
 - BAM's monetary policy stance
 - Investor expectations (interest rates and liquidity)
 - The Treasury's issuance strategy
- **Post-COVID-19**, long-term maturities have become more sensitive to shocks due to:
 - Economic and health crises
 - Inflationary pressures
 - Banking system liquidity deficits
 - Regional political instability

- Morocco is gradually adopting an **inflation targeting regime** to reinforce the credibility of its monetary policy.

- **Future research question:**

How can the implementation of an inflation targeting regime affect the cost of debt in the Moroccan context?

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Thank you for your attention!